

Scenario 1: Forecast for Next Week

You want to predict the next 7 days of website traffic. What should you do?

1. Prepare the data - Clean missing values and check for trend or seasonality.
2. Choose the right model - Use ARIMA if the data is stationary, or SARIMA if seasonality exists.
3. Generate and validate forecast - Forecast the next 7 days and review results for reasonableness.

Scenario 2: Choosing a Time Series Model

Your data has no trend and no seasonality. What model will you choose?

1. Confirm no trend and no seasonality - Plot the data and run stationarity checks (like ADF); no repeating patterns, stable mean.
2. Choose the model - Use simple models like AR or MA model
3. Fit and Test model - Fit and Test models to find which looks better and use that model

Scenario 3: Monthly Sales Data Showing Steady Growth

You observe that monthly sales data for a company has been increasing steadily over the last 2 years. How would you forecast future sales?

1. Confirm the trend - Plot monthly sales and check stationarity to verify a steady upward trend.
2. Confirm Stationarity - Do ADF test and confirm if the data is Non-Stationary
3. Convert to Stationary - Since it has a trend, apply differencing to make the data stationary.
4. Choose Model - Choose models, which can work on trend data like ARIMA
5. Generate the forecast - Fit the model and forecast future months based on the learned trend.

Scenario 4: Periodic Increase in Sales Every December

Sales data for a retail store shows a spike every December. How do you handle this while forecasting?

1. Confirm the seasonal pattern - Plot the data and verify that sales spike every December consistently.
2. Identify seasonality period - Since it's monthly data, set the seasonal cycle as 12 months.
3. Choose a seasonal model - Use SARIMA or Holt-Winters (Triple Exponential Smoothing) to explicitly model seasonality.
4. Train the model with seasonality included - Fit the model so December effects are learned, not treated as outliers.
5. Validate seasonal forecasts - Check that future December predictions show realistic spikes without distortion.

Scenario 5: Handling Missing Values in Time Series

You are analyzing a time series with missing daily values. What approach will you take?

1. Identify missing values - Check where the data is missing — randomly or in blocks.
2. Handle missing values properly - Fill gaps using forward fill, interpolation, or remove those values, depending on how many are missing.
3. Validate the result - Recheck continuity and make sure the series still makes sense before forecasting.

Scenario 6: Evaluating the Accuracy of Forecasts

You've built a time series model. How will you evaluate if it's accurate?

1. Split Data - Divide data into Train and Test set
2. Train the model - Train the model with only the Train data
3. Compare forecast with actual values - Compare forecast with actual test values
4. Calculate error metrics - Use measures like MAE, RMSE, or MAPE to quantify the forecast error.
5. Interpret the errors - Smaller error values mean better accuracy; large errors signal the model needs improvement.